

Impact of Covid-19 Pandemic on the Crude Oil and Stock Markets in the US: A time-varying analysis

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Presentation Overview

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Introduction

- **COVID-19 pandemic** has dramatically **affected global macro**.⁴
- Economic costs associated with pandemics have been substantial⁵.
- Understanding COVID-19's **impact** was crucial for **effective policy design**.
- COVID-19 brought **uncertainty** to **crude oil & stock markets**.

⁴Bloom et al., 2018

⁵**Banking & Insurance**; Leoni 2013
Govt. & Public; cummins 2006
Financial Markets; Chen et. al. 2018, Elnahas et. al. 2018



- No dearth of literature ⁶
- Past research on oil-stock relations was **often inconsistent**.⁷ Why ??



Necessity for **Unified Framework** in Analysis

- **Time Varying Complex Dynamics** between these variables. Methodology ??



Fixed Parameters, Const. variance VAR wont work !!; **TVP-VAR**⁸

⁶interaction among crude oil, stock market & macro-economic variables (Lee & Zeng, 2011; Wang & Lee, 2020).

⁷+ve correl between oil & stock; Zhu et al., 2014—→-ve correl; Narayan and Gupta ,2015

⁸Nakajima, 2011



Objective

- Model the **Dynamics** of the **COVID-19**, the **Crude Oil** & **Stock Markets** using **US Data** .
- Methodology : **TVP-VAR** (Time Varying Parameter - Vector Auto Regression)



Dataset

- **Time Series** data for the period : Jan 21, 2020 to May 6, 2020

S. No.	Variables	Proxy	Sources
1	Cov19 Pandemic	No. of Confirmed Cov19 Cases	covidtracking.com
2	Crude Oil Price	WTI Crude Oil Futures	Investing.com
3	Stock Price	S & P 500 Stock Index	Investing.com

Table: Data Sources

- 107 days = **76 Working Days** + 15 Sat & 15 Sun + 1 Holiday
- Transformation applied:
 - 1 $\ln(\# \text{ Confirmed COVID19 cases}) \Rightarrow \ln_cases$
 - 2 $\ln(\text{Stock Price}) \rightarrow 1\text{st Diff} \Rightarrow \ln_stock_diff$
 - 3 $\ln(\text{Crude Oil Price}) \rightarrow 1\text{st Diff} \Rightarrow \ln_oil_diff$



	date	cases	stock	oil
0	21-01-2020	2	3320.79	59.730
1	22-01-2020	2	3321.75	58.380
2	23-01-2020	2	3325.54	56.270
3	24-01-2020	2	3295.47	55.950
4	25-01-2020	2	3278.19	55.390
...	...	...	...	...
102	02-05-2020	1135275	2834.72	20.715
103	03-05-2020	1161034	2838.73	20.950
104	04-05-2020	1183140	2842.74	21.420
105	05-05-2020	1205532	2868.44	25.490
106	06-05-2020	1230844	2848.42	26.080

107 rows x 4 columns

(a) Raw DS

	date	ln_cases	ln_stock_diff	ln_oil_diff
0	22 / 01	0.693147	0.000289	-0.022861
1	23 / 01	0.693147	0.001140	-0.036812
2	24 / 01	0.693147	-0.009083	-0.005703
3	25 / 01	0.693147	-0.005257	-0.010059
4	26 / 01	0.693147	-0.005285	-0.010162
...	...	...	...	...
101	02 / 05	13.942385	0.001416	0.011409
102	03 / 05	13.964822	0.001414	0.011281
103	04 / 05	13.983682	0.001412	0.022186
104	05 / 05	14.002432	0.009000	0.173961
105	06 / 05	14.023211	-0.007004	0.022883

106 rows x 4 columns

(b) Processed DS

Figure: Final Dataset

- **Stationarity** is a Must for the Data !
- **Structural Breaks ??**
 - Invoke a Modified⁹ ADF test

⁹ ADF with two structural breaks; Narayan Popp, 2010



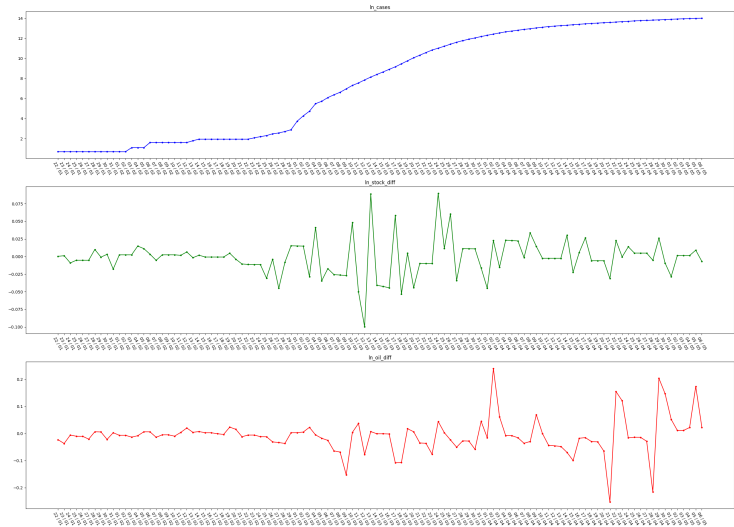


Figure: Viz. The Processed DS



Structural Breaks

- We used **PELT**¹⁰ algo. to identify the structural breaks !!
- **Two Breaks** in `ln_cases`; **No Breaks** in `ln_stock_diff` & `ln_stock_diff`



Figure: Breakpoints in `ln_cases`; `pen` = 53

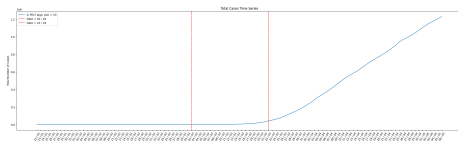


Figure: Breakpoints in Total Cases; `pen` = 53

Stationarity

- We used standard **ADF** test.
- **Non Stationary** $\rightarrow$ `ln_cases`; **Stationary** $\rightarrow$ `ln_stock_diff` & `ln_oil_diff`

```
adf_test(df["ln_cases"], autolag='AIC')
✓ 0.0s

Augmented Dickey-Fuller Test:

ADF test statistic      -1.615964
p-value                 0.474846
# lags used             7.000000
# observations           98.000000
critical value (1%)     -3.498910
critical value (5%)     -2.891516
critical value (10%)    -2.582760

Weak evidence against the null hypothesis (pval > 0.05)
Fail to reject the null hypothesis
Data has a unit root and is non-stationary
```

(a) `ln_cases`

```
adf_test(df["ln_stock_diff"], autolag='AIC')
✓ 0.0s

Augmented Dickey-Fuller Test:

ADF test statistic      -1.243667e+01
p-value                 3.833523e-23
# lags used             0.000000e+00
# observations           1.050000e+02
critical value (1%)     -3.494220e+00
critical value (5%)     -2.889485e+00
critical value (10%)    -2.581676e+00

Strong evidence against the null hypothesis (pval <= 0.05)
Reject the null hypothesis
Data has no unit root and is stationary
```

(b) `ln_stock_diff`

```
adf_test(df["ln_oil_diff"], autolag='AIC')
✓ 0.0s

Augmented Dickey-Fuller Test:

ADF test statistic      -8.804890e+00
p-value                 1.288185e-14
# lags used             0.000000e+00
# observations           1.050000e+02
critical value (1%)     -3.494220e+00
critical value (5%)     -2.889485e+00
critical value (10%)    -2.581676e+00

Strong evidence against the null hypothesis (pval <= 0.05)
Reject the null hypothesis
Data has no unit root and is stationary
```

(c) `ln_oil_diff`

Figure: The Three ADF tests



TVP - VAR Model

- we are provided a $n \times k$ time-series data.
- consider $k \times 1$ feature vectors $\tilde{y}_t$; for $t = 1, 2, \dots, s, s+1, \dots, n$
 - In our case $k = 3$ & $n = 106$
- Assume the following construct on $\tilde{y}_t$:

Time Varying Parameter - Vector Auto Regression

$$\tilde{y}_t = \tilde{c}_t + \underbrace{\mathbf{B}_{1t}\tilde{y}_{t-1} + \dots + \mathbf{B}_{st}\tilde{y}_{t-s}}_{s \text{ \# of terms [lags]}} + \tilde{e}_t; \quad \tilde{e}_t \sim N(\mathbf{0}, \mathbf{\Omega}_t); \text{ for } t = s+1, s+2, \dots, n \quad (1)$$

- 1 $\mathbf{\Omega}_t \rightarrow k \times k$ Time Varying Covariance Matrix
- 2 $\mathbf{B}_{it} \rightarrow k \times k$ Time Varying Coefficients Matrix; for $i = 1, 2, \dots, s$

- **$s = 1$ is optimal for our case**; based on *Infomation Criteria* : AIC, HQIC, & BIC.



Optimal Lag

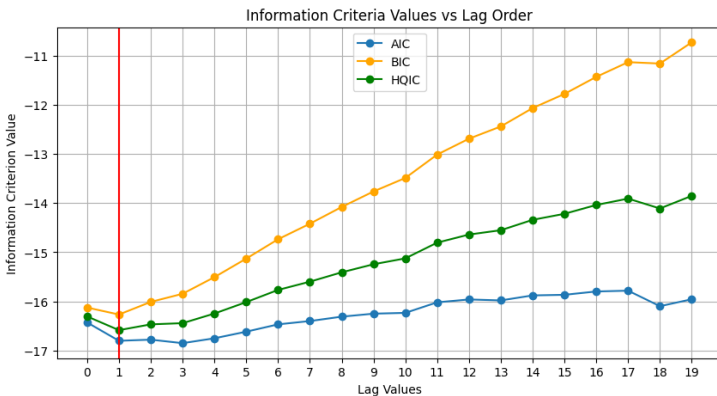


Figure: Selecting optimal lag, s ; using AIC, BIC, HQIC



- Defining Ω_t for $t = s + 1, s + 2, \dots, n$; in a **recursive identification** fashion

$$\boxed{\Omega_t = \mathbf{A}_t^{-1} \Sigma_t \Sigma_t' \mathbf{A}_t'^{-1}} \equiv (\mathbf{A}_t^{-1} \Sigma_t)(\mathbf{A}_t^{-1} \Sigma_t)' \quad (2a)$$

$$\underbrace{\Sigma_t}_{k \times k \text{ diagonal matrix}} = \text{diag}(\sigma_{1t}, \sigma_{2t}, \dots, \sigma_{kt}) \quad (2b)$$

$$\underbrace{\mathbf{A}_t}_{k \times k \text{ lower } \triangle \text{ matrix}} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ a_{1t} & 1 & 0 & \dots & 0 & 0 \\ a_{2t} & a_{3t} & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \dots & a_{qt} & 1 \end{pmatrix} \quad (2c)$$

$$\text{where } q = \frac{k(k-1)}{2}$$

- Further we will define the following parameters:

$$\underbrace{\beta_t}_{\text{stacked row vector of } \mathbf{B}_{it}} = \boxed{(\mathbf{B}_{1t}, \mathbf{B}_{2t}, \dots, \mathbf{B}_{st})} \quad (3a)$$

$$\tilde{h}_t = (h_{1t}, h_{2t}, \dots, h_{kt}); \text{ where } h_{it} = \log(\sigma_{it}^2) \text{ for } i = 1, 2, \dots, k \quad (3b)$$

$$\underbrace{\underline{a}_t}_{\text{stacked row vector of } \mathbf{A}_t} = (a_{1t}, a_{2t}, \dots, a_{qt})' \quad (3c)$$



- We will *assume* that these TVPs follow **Random Walk Process**

$$\tilde{\beta}_{t+1} = \tilde{\beta}_t + \underline{u}_{\beta t} \quad (4a)$$

$$\tilde{a}_{t+1} = \tilde{a}_t + \underline{u}_{a t} \quad (4b)$$

$$\tilde{h}_{t+1} = \tilde{h}_t + \underline{u}_{h t} \quad (4c)$$

$$\begin{pmatrix} \tilde{\varepsilon}_t \\ \underline{u}_{\beta t} \\ \underline{u}_{a t} \\ \underline{u}_{h t} \end{pmatrix} \sim N \left(0, \begin{pmatrix} \underline{I} & \underline{O} & \underline{O} & \underline{O} \\ \underline{O} & \underline{\Sigma}_{\beta} & \underline{O} & \underline{O} \\ \underline{O} & \underline{O} & \underline{\Sigma}_a & \underline{O} \\ \underline{O} & \underline{O} & \underline{O} & \underline{\Sigma}_h \end{pmatrix} \right) \quad (4d)$$

where $t = s + 1, s + 2, \dots, n - 1, n$

- Further

$$\tilde{\beta}_{s+1} \sim N(\underline{\mu}_{\beta 0}, \underline{\Sigma}_{\beta 0}) \quad (5a)$$

$$\tilde{a}_{s+1} \sim N(\underline{\mu}_{a 0}, \underline{\Sigma}_{a 0}) \quad (5b)$$

$$\tilde{h}_{s+1} \sim N(\underline{\mu}_{h 0}, \underline{\Sigma}_{h 0}) \quad (5c)$$

$$\underline{e}_t = (\underline{A}_t^{-1} \underline{\Sigma}_t) \underline{\varepsilon}_t \quad (5d)$$

- $\underline{\Sigma}_a$ & $\underline{\Sigma}_h \rightarrow$ are diagonal Matrix
- $\underline{\Sigma}_{\beta}$???



- Assume the following apriori

$$\begin{aligned}\underline{\mu}_{\beta 0} &= \underline{\mu}_{a0} = \underline{\mu}_{h0} = \underline{0} \\ (\Sigma_{\beta 0})_i^{-2} &\sim \Gamma(20, 10^{-4}) \\ (\Sigma_{a0})_i^{-2} &\sim \Gamma(4, 10^{-4}) \\ (\Sigma_{h0})_i^{-2} &\sim \Gamma(4, 10^{-4})\end{aligned}$$

- $(\cdot)_i \rightarrow i^{th}$ diagonal element
- Computing the posterior dist. of parameters:
 - the MCMC algorithm iterated 10,000 times, discarding the initial 1,000 samples.



- **Bayesian MCMC** used for TVP-VAR model
- Estimating **posterior distribution** of the parameters

$$\underbrace{p(\theta | x)}_{\text{Posterior Probability}} = \frac{p(x | \theta)}{\underbrace{p(x)}_{\text{Data}}} p(\theta)$$

- **Gibbs Sampler**: cdl. distribution manageable but not joint distribution
- **Metropolis-Hastings**: posterior distribution manageable but not cdl. distribution Algo.
- **Geweke diagnostic** - Convergence of Markov Chain



- Estimation result in the TVP-VAR Model

Parameter	Mean	Stdev	95%L	95%U	Geweke	Inef.
sb1	0.0027	0.0005	0.0019	0.0039	0.084	23.19
sb2	0.0027	0.0004	0.0021	0.0036	0.549	17.18
sa1	0.0057	0.0020	0.0034	0.0100	0.755	33.01
sa2	0.0054	0.0016	0.0033	0.0091	0.387	24.84
sh1	1.2280	0.2003	0.8348	1.6433	0.142	197.23
sh2	0.3336	0.0708	0.1946	0.4663	0.000	251.33

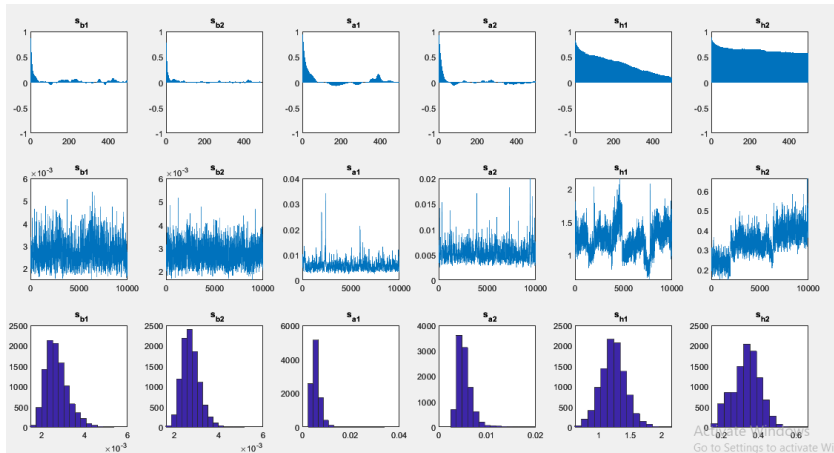
TVP-VAR model (Lag = 1)

Iteration: 10000



Results

- The first, second and third row show **sample autocorrelations**, **sample paths**, **posterior densities**, respectively

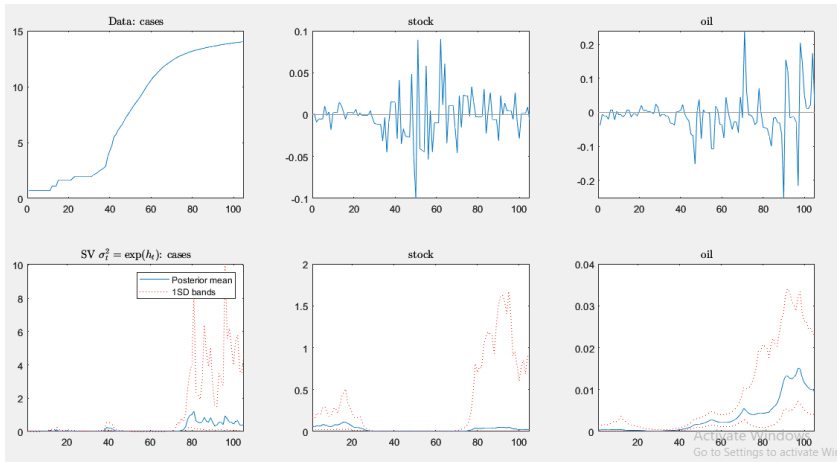


Activate Windows
Go to Settings to activate Windows.



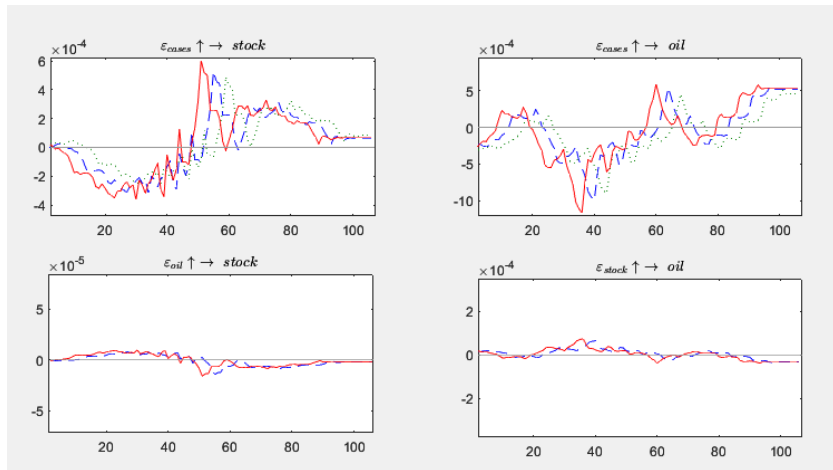
Results

- Sample Trajectory



Results

- **Impulse Responses** - 1% Unit Impulse



Conclusion

- 1 **-ve correlation** between **crude oil** & **stock returns**.
- 2 Surprisingly, **COVID-19 pandemic** has a statistically significant **+ve impact** on both **crude oil** & **stock returns**.



- COVID-19 **may not** harm economic performance, **at least** in the timeframe of our study.
- **Short-term policy** interventions **may not be required** for the pandemic's impact on crude oil & stock markets.



References



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End

THANKS !

